POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

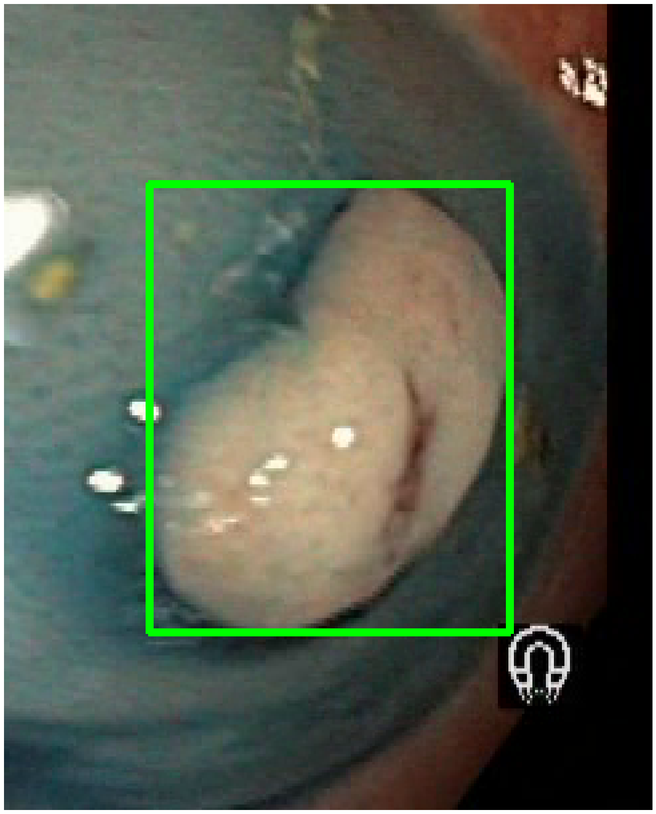
Video ID:	6822e0ee-9ee2-4408-8c55-3c1030928679
Total Frames:	4642
Frame Rate:	25.0 fps (assumed)
Duration:	185.7 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	3719	✓
RT-DETR Detections	3941	✓
MedSAM2 Detections	4026	✓
Consensus (3-Model Agreement)	5	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	2	40.0%
MEDIUM RISK	0	0.0%
LOW RISK	3	60.0%
TOTAL ANALYZED	5	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 39 frames)



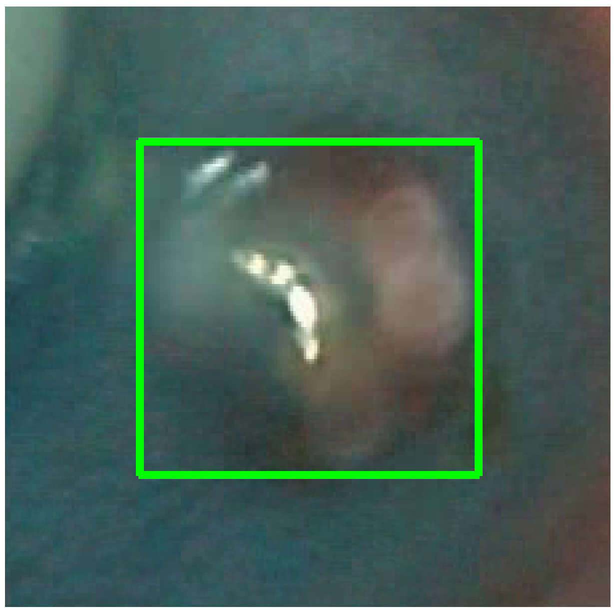
Feature	Value	Interpretation
Frame Index	3775	Frame 3775 of 4642
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 74.7%
Redness Score	0.020	Color-based vascularization
Relative Radius	26.0%	Polyp size as % of frame diagonal
Texture Score	0.481	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket

Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	94.4%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 1535 frames)



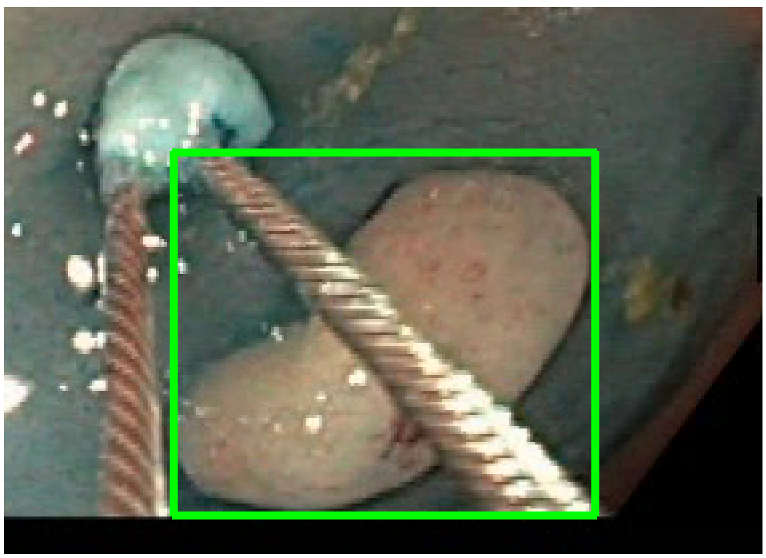
Feature	Value	Interpretation
Frame Index	2738	Frame 2738 of 4642
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 75.8%
Redness Score	0.002	Color-based vascularization
Relative Radius	38.3%	Polyp size as % of frame diagonal
Texture Score	0.409	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	91.1%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 458 frames)



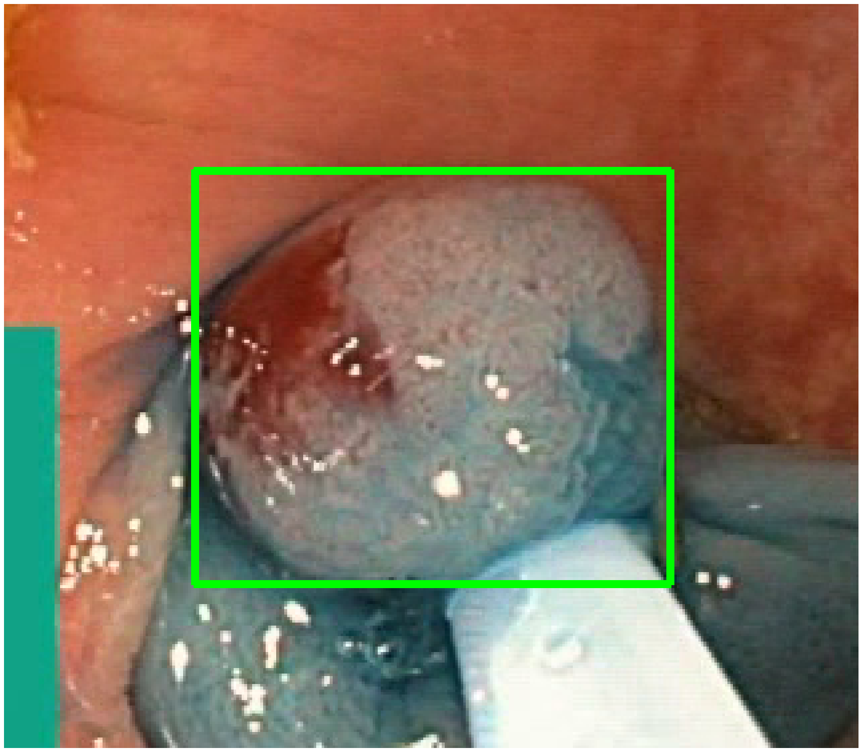
Feature	Value	Interpretation
Frame Index	3854	Frame 3854 of 4642
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**LIFTED_POLYP**	Type confidence: 71.7%
Redness Score	0.044	Color-based vascularization
Relative Radius	10.3%	Polyp size as % of frame diagonal
Texture Score	0.349	Surface morphology
Vessel Visibility	0.004	Vascularization indicator

Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	92.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 1355 frames)



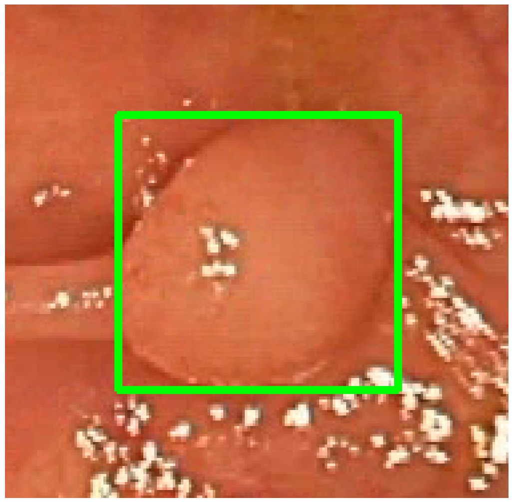
Feature	Value	Interpretation
Frame Index	4385	Frame 4385 of 4642
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.256	Color-based vascularization
Relative Radius	21.0%	Polyp size as % of frame diagonal
Texture Score	0.336	Surface morphology
Vessel Visibility	0.706	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended

Detection Confidence	92.2%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #5 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 38 frames)



Feature	Value	Interpretation
Frame Index	2123	Frame 2123 of 4642
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**BLEEDING_POLYP**	Type confidence: 92.0%
Redness Score	0.263	Color-based vascularization
Relative Radius	11.0%	Polyp size as % of frame diagonal
Texture Score	0.399	Surface morphology
Vessel Visibility	0.954	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_POLYP	High suspicion for malignancy — biopsy and urgent resection recommended

Detection Confidence	91.5%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:
A total of 5 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 2 (40.0%)
• MEDIUM RISK polyps: 0 (0.0%)
• LOW RISK polyps: 3 (60.0%)

Recommendation:
The presence of HIGH RISK polyps warrants close clinical attention. Consider endoscopic resection or follow-up surveillance depending on clinical context.

Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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